Dr. Ivan Sergio
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Education

Fisciano, Italy	University of Salerno
Nov 2019 – Sep 2023	Ph.D in Public Economics
Fisciano, Italy	University of Salerno
Sep 2016 – Sep 2019	Master's Degree in Political Science
Foggia, Italy	University of Foggia
Sep 2013 – Sep 2016	Bachelor's Degree in Economics

Research Experience

Hannover, Germany — Leibniz FH
Dec 2023 – Present — Ph.D in Public Economics

- Conducting empirical research in public economics, regional development and economic geography, with a focus on innovation, productivity and technological change.
- Applying advanced econometric and quantitative methods, including panel data analysis, spatial econometrics and machine learning.
- Preparing scientific publications and contributing to research projects and academic collaborations.

Hamburg, Germany — Helmut Schimdt University
Oct 2023 – Sep 2026 — Spatial Economics group in collaboration with HWWI

- Conducting quantitative research on regional economic development, spatial inequalities and technological innovation.
- Developing machine-learning models to reconstruct historical regional GDP and investigate long-term patterns of European economic development.
- Applying spatial econometric methods, GIS and panel data techniques to analyse regional productivity, green economic transformation and cryptocurrency adoption.

Santiago de Compostela, Spain — AYeconomics
Sep 2022 – Dec 2022 — Researcher

- Conducting quantitative research on regional economic development, spatial inequalities and technological innovation.
- Preparing scientific publications and contributing to research projects and academic collaborations.

Fisciano, Germany — University of Salerno
Nov 2019 – Sep 2023 — Researcher

- Conducted doctoral research on ICT innovation, green productivity and industrial agglomeration.
- Investigated the relationship between technological innovation, regional economic performance and environmental sustainability using quantitative and econometric methods.
- Presented research findings at international conferences and contributed to scientific publications.

Research Project

Hamburg, Germany

2024 – Present "The Origins of Modern European Economic Geography"

- Developed a machine-learning framework to reconstruct historical GDP per capita across European NUTS2 regions over more than a millennium.
- Integrated historical datasets on urbanisation, transportation infrastructure, population and economic activity, applying Random Forest and XGBoost algorithms.
- Evaluated the accuracy and robustness of historical regional GDP estimates against existing economic history datasets and investigated long-term patterns of regional economic divergence and convergence.

Hamburg, Germany

2023 – Present "Green Energy Transition and European Regional Productivity" (hyBit and Greengo projects included)

- Applied the Malmquist-Luenberger productivity index to measure regional productivity changes while accounting for undesirable environmental outputs.
- Used panel econometric and instrumental variable techniques to investigate the effects of energy composition on regional productivity and environmental performance.
- Conducted quantitative research on energy efficiency, productivity and the role of hydrogen in the transition towards sustainable industrial production.
- Contributed to scientific publications and research activities examining the implications of hydrogen-based industrial transformation.

Fisciano, Italy

2019 – 2023 "ICT Innovation, Industrial Agglomeration and Regional Economic Performance"

- Investigated the relationship between ICT innovation, industrial agglomeration and economic performance, focusing on the role of technological development in regional and industrial transformation.
- Produced two scientific publications contributing to the literature on ICT innovation, industrial dynamics and economic performance.

Fisciano, Italy

2019 – 2023 "R&D Investment and Firm Growth: Evidence from ICT Small and Medium-Sized Enterprises"

- Investigated the determinants of firm growth among ICT small and medium-sized enterprises (SMEs), with particular attention to the role of research and development (R&D) investment.
- Produced two scientific publications contributing to the literature on ICT innovation, industrial dynamics and economic performance.

Publication (peer-reviewd)

- **Sergio I.**, The Origins of Modern European Economic Geography: Evidence from a Machine Learning Reconstruction of Regional GDP, 700-2000 (2026). Available at SSRN: https://ssrn.com/abstract=6341378 or http://dx.doi.org/10.2139/ssrn.634137
- D'Errico M.C., Ferragina A.M., Iandolo S., **Sergio I.**, Beyond productivity: Does ICT agglomeration drive regional eco-efficiency?, *Structural Change and Economic Dynamics*, 2026, https://doi.org/10.1016/j.strueco.2026.09.009
- Bogdzinski H., **Sergio I.**, Wedemeier J., Unveiling the impact of green energy on green productivity: A focus on hydrogen and the green transition in European regions, *Economic Analysis and Policy*, Volume 86, 2025, Pages 1064-1082,ISSN 0313-5926, https://doi.org/10.1016/j.eap.2025.03.044.
- **Sergio I.,** Iandolo S., Ferragina A.M., Inter-sectoral and inter-regional knowledge spillovers: The role of ICT and technological branching on innovation in high-tech sectors,*Technological Forecasting and Social Change*, Volume 194, 2023, 122728, ISSN 0040-1625, https://doi.org/10.1016/j.techfore.2023.122728.
- **Sergio, I.**, Wedemeier, J. Global surge: exploring cryptocurrency adoption with evidence from spatial models. *Financ Innov* 11, 96 (2025).
- Bernhardt, L. & **Sergio, I.** (2025). Konjunkturschlaglicht: Kupfer als Frühindikator wirtschaftlicher Entwicklung in Deutschland?. *Wirtschaftsdienst*, 105(8), 2025. 607-608. https://doi.org/10.2478/wd-2025-0153

- Ferragina, A. M., Iandolo, S., **Sergio, I.**, & Taymaz, E. (2024). Digital economy and environmental performances in the MENA area. In S. Capasso & G. Canitano (Eds.), *Mediterranean Economies 2024: The new agenda for the Mediterranean—Perspectives and challenges*. Bologna: Il Mulino. https://www.darwinbooks.it/doi/10.978.8815/413758
- **Sergio, I.**, & Petti, D. (2024). The impact of geopolitical risks on Bitcoin volume growth: Evidence from a panel data analysis. arXiv. https://arxiv.org/abs/2409.17057
- Ferragina, A. M., Iandolo, S., **Sergio, I.**, & Taymaz, E. (2024). Digital economy and environmental performances in the MENA area. In S. Capasso & G. Canitano (Eds.), *Mediterranean Economies 2024: The new agenda for the Mediterranean—Perspectives and challenges*. Bologna: Il Mulino. https://www.darwinbooks.it/doi/10.978.8815/413758
- Ferragina, A. M., Iandolo, S., **Sergio, I.**, & Taymaz, E. (2023, September 27–29). ICT as a driver for green economic growth in the MENA area. In *FEMISE ERF Conference 2023: Shifting paradigms—Opportunities for a deeper EU-Mediterranean integration in a changing world* (Sala de Actes, European Institute of the Mediterranean [IEMed], Barcelona, Spain).
- Peña-Boquete, Y., **Sergio, I.**, & Seredenko, M. (2023). Economic consequences for Spain and Italy of export disruption to Russia. In S. Capasso & G. Canitano (Eds.), *Mediterranean Economies 2023: The impact of the Russia-Ukraine war in the Mediterranean region—The socio-economic consequences* (digital ed.). Bologna: Il Mulino. https://www.darwinbooks.it/doi/10.978.8815/411167

Working paper

- **Sergio I.**, Ferragina A.M., Iandolo S., The impact of R&D on the information and communication technology (ICT) firms' growth. A firm level analysis on EU countries. (advanced stage)
- Ferragina A.M., Baylie L, Iandolo S., **Sergio I.**, Environmental Regulation and Trade Performance: Evidence from Italian Provinces using DiD (advanced stage)
- **Sergio I.**, Bourdin S., The Evolution of Regional Inequalities in Europe, a Focus on Institutional Stability

Converence participation

as keynote

- **Ivan Sergio**, "The role of ICT in a modern world", Diginomics, Bremen, June 2025

as organizer

- Italian Trade Study Group, University of Salerno, Salerno, Italy, 2022

as presenter

- **Sergio I.**, "The Origins of Modern European Economic Geography: Evidence from a Machine Learning Reconstruction of Regional GDP, 700-2000", presentetd at Associazione Italiana Scienze Regionali (AISRE) 2026, Florence, Italy, 25 June 2026
- Bogdzinski H., **Sergio I.**, Wedemeier J., "Unveiling the impact of green energy on green productivity: A focus on hydrogen and the green transition in European regions", presented at Società Italiana di Economia (SIE), Naples, Italy, 2025
- D'Errico M.C., Ferragina A.M., Stefano Iandolo, **Ivan Sergio**, "Beyond productivity: Does ICT agglomeration drive regional eco-efficiency?", presented at Associazione Italiana Scienze Regionali (AISRE) 2026, Torino, Italy, 2024
- **Sergio I.**, Iandolo S., Ferragina A.M., "Inter-sectoral and inter-regional knowledge spillovers: The role of ICT and technological branching on innovation in high-tech sectors", presented at Società Italiana di Economia (SIE), L'Aquila, Italy, 2023
- **Sergio I.**, Iandolo S., Ferragina A.M., "Inter-sectoral and inter-regional knowledge spillovers: The role of ICT and technological branching on innovation in high-tech sectors", presented at European Regional Science Association (ERSA) , Pécs, Bulgary, 2023
- **Sergio I.**, Iandolo S., Ferragina A.M., "The role of ICT and technological branching on innovation in high-tech sectors", presented at GAME, Santiago de Compostela, Spain, 2022
- **Sergio I.**, Iandolo S., Ferragina A.M., "The impact of R&D on the information and communication technology (ICT) firms' growth. A firm level analysis on EU countries.", presented at Società Italiana di Economia (SIE), Online, Italy, 2022

Post-Graduate course

- **London School of Economics Masterclass.** (Madrid, 2025). The wealth and Poverty of regions: innovation, economic growth and the challenge of the 'places left behind'. (Madrid, 2025) LSE & Fundaciòn Ramòn Areces – Professors: Andrès Rodriguez-Pose & Neil Lee
- **AISRE Summer School.** (Rome, 2024). Policy Evaluation in Regional Science. Rome
- **Spatial Econometrics Advanced Institute.** (Online, 2023) .Summer School at the SEA with Prof. J. Paul Elhorst
- **Course on input-output introduction.** (Online, 2023) AYeconomics Research Centre
- **Financial Time Series and High Frequency Econometrics**. (Online, 2020) Italian Econometric Association

Awards

2013 – 2015	Merit Scholarship, Academic Excellence Award ~ EUR 5.000 per year
2019 – 2023	Awarded a total of €3,000 in university scholarships (University of Salerno) to provide financial support during doctoral studies.

Teaching

Apr 2026 – Sep 2026	Lecture of "Markets and Public Policy" - „Markt und Staat" -, Leibiniz FH
Apr 2020 – Sep 2024	Teacher assistant of "International Economics", University of Salerno

Skills

Econometrics & Quantitative Methods	Panel Data Econometrics, Spatial Econometrics, Instrumental Variable (IV) Estimation, Fixed Effects Models, Causal Inference, Productivity Analysis (DEA, Malmquist Index), Statistical Modelling.
Data Science & Machine Learning	R (Tidyverse, ggplot2, sf), Machine Learning (Random Forest, XGBoost), Data Processing, Predictive Modelling and Data Visualisation.
GIS & Spatial Analysis	QGIS, Geographic Information Systems (GIS), Spatial Data Management, Regional Economic Analysis, Spatial Weight Matrices, Geospatial Visualisation.
Software & Research Tools	Stata (reghdfe, ivreghdfe, spreg, xsmle), RStudio, Python, LaTeX, Microsoft Office.
Research & Project Management	Scientific Writing, Grant and Research Proposal Writing, Academic Publishing, Project Management, Research Coordination, Interdisciplinary Collaboration.
Languages	Italian (Native), English (Fluent), Spanish (Intermediate), German (Basic).

Grants & Research Funding

Acquired funding (in €)	*Funding agency*	*Project duration*	*Project title (incl. link to project webpag*	*My role in this project*
56.816,00	MIUR Funding	4 years	*Greengo*	Researcher

30.000.000	Horizon 2020	3.5 years	*hyBit– Link: https://www.uni-bremen.de/res/forschung/hybit*	Researcher

Interest

Tennis, Naples History, South Question, GIS, Causal inference

Contact reference

Prof. Dr. Claudie Diebolt – University of Strasburg - cdiebolt@unistra.fr
Prof. Dr. Sébastien Bourdin - IESEG School Of Management - s.bourdin@ieseg.fr
Prof. Dr. Anna Maria Ferragina – University of Salerno – aferragina@unisa.it

I hereby give consent for my personal data to be processed and stored for recruitment purposes. This is in accordance with the General Data Protection Regulation (EU) 2016/679.